STUDY BUDDY STEM AI: AN AI-POWERED PERSONALIZED, ADAPTIVE AND MULTIMODAL STEM LEARNING ECOSYSTEM FOR SRI LANKAN STUDENTS

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Individual component: Syllabus-Grounded Multilingual Conversational AI Voice Assistant for Interactive STEM Tutoring

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Artificial intelligence-based educational systems provide opportunities for students to obtain immediate and interactive learning support; however, general-purpose conversational models may generate responses that are not aligned with a specific curriculum and may produce unsupported or inaccurate information. These limitations are particularly important in school-level education, where responses should remain consistent with prescribed learning materials. This research addresses the problem through the development of a syllabus-grounded multilingual conversational artificial intelligence voice assistant for Sri Lankan G.C.E. Ordinary Level Science education. The proposed component forms part of the Study_Buddy intelligent STEM tutoring ecosystem and is designed for Grade 10 and Grade 11 students. The system combines browser-based speech capture, automatic speech recognition, retrieval-augmented generation, semantic vector retrieval, conversational response generation and text-to-speech synthesis. The knowledge base is constructed from National Institute of Education Science textbooks and Teacher's Guides and is indexed using semantic embeddings and FAISS similarity search. The speech-recognition layer uses a locally executable Faster-Whisper configuration, while cloud and local language-model alternatives are incorporated to improve operational flexibility. English and Sinhala interaction are supported through language-aware processing and response generation. A controlled prompting strategy is used to constrain generated responses to retrieved curriculum evidence and to provide an explicit refusal when sufficient syllabus-grounded evidence is unavailable. The implemented component is integrated with the wider Study_Buddy platform through REST-based FastAPI endpoints. Functional and integration testing are used to verify the end-to-end interaction pipeline. Quantitative speech-recognition, retrieval and user-validation measures are reported separately after completion of the final evaluation. The research demonstrates a practical architecture for combining voice interaction, curriculum-grounded retrieval and multilingual conversational tutoring within a resource-conscious educational environment.

Keywords: Retrieval-Augmented Generation, Voice Tutor, Sinhala Speech Recognition, STEM Education, Conversational AI

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
CER	Character Error Rate
CPU	Central Processing Unit
FAISS	Facebook AI Similarity Search
G.C.E.	General Certificate of Education
LLM	Large Language Model
NIE	National Institute of Education
NLP	Natural Language Processing
O/L	Ordinary Level
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
STT	Speech-to-Text
TTS	Text-to-Speech
UI	User Interface
VAD	Voice Activity Detection
WER	Word Error Rate

1 INTRODUCTION

1.1 Overall project context

Artificial intelligence has increasingly been incorporated into educational technologies to provide personalized learning support, automated feedback and interactive access to educational resources. Conversational artificial intelligence is particularly relevant because it enables learners to communicate with educational systems using natural-language questions rather than relying exclusively on conventional menus, search interfaces or predefined learning paths.

The Study_Buddy project was developed as an intelligent STEM tutoring ecosystem for Sri Lankan G.C.E. Ordinary Level Science education. The system focuses on Grade 10 and Grade 11 learners and seeks to provide curriculum-oriented learning support through multiple intelligent learning components. The overall ecosystem incorporates complementary capabilities including conversational voice tutoring, knowledge mapping, adaptive assessment and narrative learning.

The individual research presented in this thesis focuses on the Voice Tutor component. Rather than treating artificial intelligence as a general-purpose question-answering mechanism, the component is designed around the principle that educational responses should be grounded in authoritative curriculum resources. This distinction is important because a response that is generally factually correct may still be inappropriate if it conflicts with the terminology, depth or scope expected within the Sri Lankan school curriculum.

The proposed Voice Tutor therefore combines speech interaction with curriculum-grounded retrieval. Student speech is converted into text through automatic speech recognition, relevant sections of the approved knowledge base are retrieved using semantic similarity, and the retrieved material is supplied to a conversational language model to generate an educational response. The response can subsequently be converted into speech, enabling an end-to-end voice interaction.

The use of a curriculum-specific knowledge base also supports a controlled response strategy. When relevant evidence cannot be retrieved, the system can avoid presenting an unsupported generated answer and instead communicate that the requested information is outside the available syllabus resources.

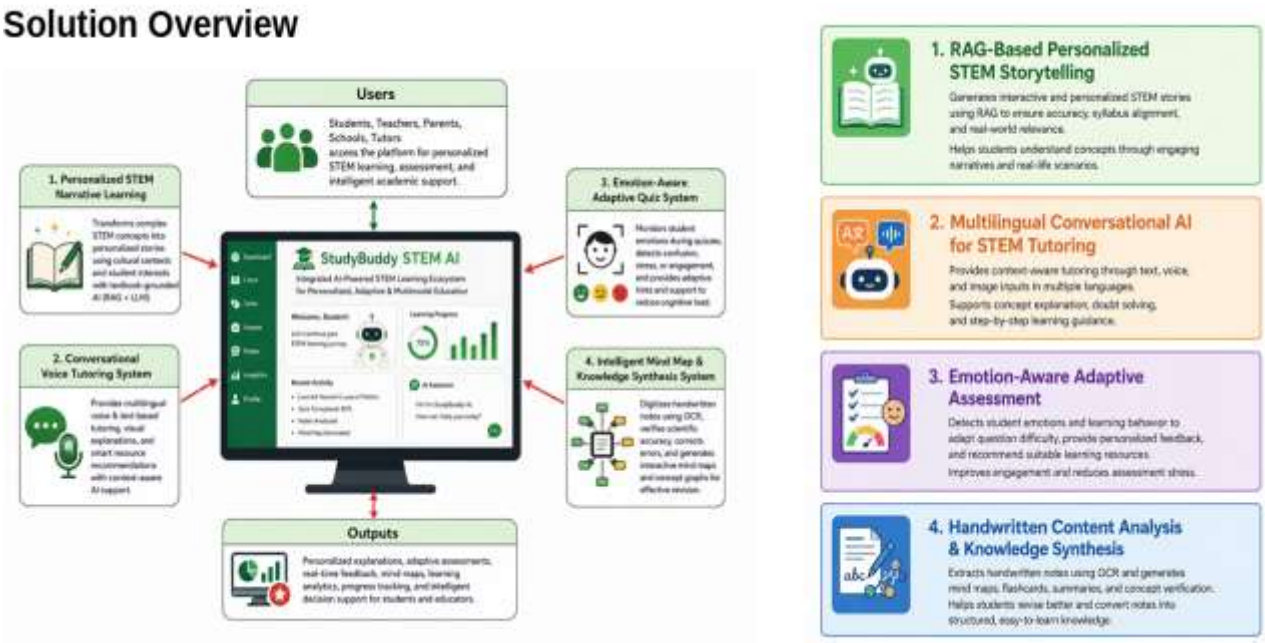


Figure 1 - System Overview

1.2 Individual component context

The individual component is titled Syllabus-Grounded Multilingual Conversational AI Voice Assistant for Interactive STEM Tutoring.

The component is designed to provide students with a natural voice-based interface for asking questions related to Grade 10 and Grade 11 Science. The system supports English and Sinhala interaction and is designed to operate as part of the broader Study_Buddy ecosystem.

The component consists of five major processing stages:

1. speech capture;
2. automatic speech recognition;

3. curriculum-grounded retrieval;
4. conversational response generation; and
5. text-to-speech output.



Figure 2 - Individual component overview

The browser captures student speech using the Media Recorder interface. The recorded audio is submitted to the Fast API backend, where the local speech-recognition pipeline converts the recording into text. The resulting question is then processed by the retrieval component.

The curriculum knowledge base is constructed from Grade 10 and Grade 11 Science textbooks and Teacher's Guides published by the National Institute of Education. Documents are extracted, segmented into manageable text chunks and represented using semantic embeddings. FAISS is used to identify the most semantically relevant chunks for a student's question.

The retrieved context is supplied to a language model with instructions to behave as a Sri Lankan O/L Science tutor. The generated response is expected to remain within the retrieved curriculum context. The resulting answer can be returned as text and converted into speech using a text-to-speech service.

1.3 Problem statement

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The retrieved context is supplied to a language model with instructions to behave as a Sri Lankan O/L Science tutor. The generated response is expected to remain within the retrieved curriculum context. The resulting answer can be returned as text and converted into speech using a text-to-speech service.

1.3 Problem statement

Students studying school-level Science may require immediate clarification of concepts, terminology and relationships between topics. Conventional educational resources provide authoritative information but do not necessarily provide an interactive conversational mechanism through which students can ask questions in their preferred language.

General-purpose conversational AI systems offer natural interaction, but their broad knowledge scope introduces a significant educational concern: a model may generate information that is technically plausible but not supported by the prescribed curriculum. In a school examination context, curriculum alignment is important because students are expected to understand concepts according to the knowledge, terminology and depth established by the relevant educational materials.

A second challenge concerns accessibility of interaction. Text-based question-answering systems require students to formulate written queries. A voice interface can provide a more natural interaction mechanism, particularly for exploratory questions and conversational follow-up.

A third challenge is the Sinhala language context. Sinhala is a comparatively low-resource language for speech technology, creating additional considerations for accurate speech recognition and multilingual educational interaction. Recent research on Sinhala ASR demonstrates active work on improving speech recognition using multilingual pretrained models, while also illustrating the importance of evaluation in low-resource language settings [1].

Therefore, the research problem addressed by this thesis is:

How can a multilingual conversational voice assistant be designed and implemented to provide curriculum-grounded, interactive and reliable Science tutoring for Sri Lankan G.C.E. O/L students while reducing dependence on unsupported generative responses?

1.4 Research gap

Existing research demonstrates the potential of retrieval-augmented generation for educational question answering. Levonian et al. investigated RAG for middle-school mathematics and found that curriculum-grounded responses can improve educational question-answering, while also identifying a trade-off between strict grounding and learner preference [2]. Alawwad et al. similarly demonstrated the value of combining retrieval with language models for textbook question answering [3].

Research has also investigated multilingual and Sinhala speech recognition. Harischandra et al. evaluated pretrained multilingual models for low-resource Sinhala ASR, demonstrating the relevance of transfer learning for Sinhala speech processing [1].

However, these areas are often investigated independently. There remains a practical research opportunity to combine:

- voice-based interaction;
- Sinhala-English multilingual support;
- curriculum-grounded retrieval;
- conversational response generation;
- local speech recognition;
- controlled out-of-syllabus behaviour; and
- resource-conscious deployment

within a single educational tutoring component specifically oriented towards Sri Lankan O/L Science.

The research therefore addresses a combined technological and contextual gap rather than claiming that individual techniques such as RAG, ASR or LLMs are themselves novel.

1.5 Aim and objectives

Aim

The aim of this research is to design, implement and evaluate a syllabus-grounded multilingual conversational AI voice assistant that provides interactive Science tutoring for Sri Lankan G.C.E. O/L students using curriculum-based retrieval and speech technologies.

Objectives

The objectives of the research are:

O1: To investigate existing approaches to voice-based educational assistants, retrieval-augmented generation, multilingual speech recognition and curriculum-grounded question answering.

O2: To design a curriculum-grounded knowledge retrieval architecture using official Grade 10 and Grade 11 Science educational resources.

O3: To implement an automatic speech-recognition pipeline capable of processing student voice queries.

O4: To implement semantic retrieval using embedding-based similarity search and FAISS.

O5: To integrate conversational language-model generation with retrieved curriculum context.

O6: To support English and Sinhala conversational responses.

O7: To implement text-to-speech functionality for conversational educational interaction.

O8: To evaluate the functional correctness, retrieval behavior, speech-processing performance and usability of the developed component.

O9: To integrate the Voice Tutor with the wider Study_Buddy platform.

Table 1.1 should be used to map each objective to its corresponding implementation and evaluation evidence.

1.6 Research questions or hypotheses

The study investigates the following research questions:

RQ1: How can retrieval-augmented generation be used to constrain conversational AI responses to Sri Lankan O/L Science curriculum resources?

RQ2: How effectively can a local automatic speech-recognition pipeline support English and Sinhala voice interaction within a school-level educational assistant?

RQ3: How can semantic vector retrieval be incorporated into a conversational voice-tutoring architecture to provide relevant curriculum context?

RQ4: What technical and usability considerations arise when integrating speech recognition, curriculum retrieval, language-model generation and text-to-speech into a unified educational system?

RQ5: To what extent does the developed component satisfy the functional and educational requirements defined for the Study_Buddy platform?

1.7 Scope limitations and assumptions

The research focuses on the development of a Voice Tutor for Sri Lankan G.C.E. O/L Science education.

The scope includes:

- Grade 10 and Grade 11 Science;
- English and Sinhala interaction;
- speech-to-text processing;
- curriculum-based semantic retrieval;
- conversational answer generation;
- text-to-speech output;

- curriculum-grounded refusal behavior.
- Fast API-based backend services;
- integration with the wider Study Buddy system.

The research does not attempt to:

- replace qualified teachers;
- provide unrestricted general-purpose medical, legal or scientific advice;
- cover every Sri Lankan school subject;
- guarantee perfect speech recognition;
- establish clinical or psychological educational outcomes;
- claim national-scale educational effectiveness without a sufficiently large validation study.

The system assumes that the selected curriculum documents accurately represent the intended knowledge base and that the final evaluation environment provides suitable audio recordings and testing conditions.

1.8 Individual contribution

The individual contribution of this research consists of the design and implementation of the Voice Tutor component within Study Buddy .

The contribution includes:

- investigation of conversational educational AI;
- design of the speech interaction pipeline;
- implementation of speech capture and transcription;
- preparation of curriculum-based knowledge resources;
- implementation of semantic embeddings;
- implementation of FAISS-based retrieval;
- design of curriculum-grounded prompting;
- integration of language-model generation;
- implementation of multilingual response handling;

- integration of text-to-speech;
- API development;
- error-handling and fallback mechanisms;
- functional testing;
- evaluation planning and analysis; and
- integration with the other project components.

The individual component therefore represents a complete research and development contribution rather than only a software module.

1.9 Thesis structure

Chapter 1 introduces the research context, problem, research gap, objectives, research questions, scope and individual contribution.

Chapter 2 reviews the theoretical and technological foundations relevant to conversational AI tutoring, automatic speech recognition, retrieval-augmented generation, semantic retrieval and multilingual educational systems.

Chapter 3 presents research methodology, study materials, evaluation procedures, metrics, data-analysis approach and responsible-AI considerations.

Chapter 4 describes the product and solution design, including requirements, architecture, design alternatives and detailed component design.

Chapter 5 explains implementation, integration, software testing and deployment.

Chapter 6 presents the research and technical results obtained from the evaluation.

Chapter 7 interprets the findings, compares them with related work and discusses research and product contributions, limitations and implications.

Chapter 8 discusses commercialization and sustainability.

Chapter 9 concludes the thesis and identifies future development directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review focuses on research directly relevant to the development of the proposed Voice Tutor component. The main themes include:

1. Artificial intelligence in education;
2. Conversational educational systems;
3. Retrieval-Augmented Generation (RAG);
4. Textbook and curriculum question answering;
5. Automatic Speech Recognition (ASR);
6. Low-resource Sinhala speech recognition;
7. Semantic embeddings and vector retrieval;
8. Multilingual conversational systems; and
9. Responsible use of generative AI in education.

The review prioritises peer-reviewed publications, established technical papers, official documentation, and authoritative curriculum resources. These sources provide the theoretical and technical foundations for the proposed system and help identify limitations in existing approaches.

The literature is used to establish the rationale for integrating curriculum-grounded question answering, semantic retrieval, speech recognition, language generation, and speech synthesis within a single educational interaction workflow.

2.2 Domain and theoretical foundations

The literature review focuses on research directly relevant to the development of the proposed Voice Tutor component. The main themes include:

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The literature is used to establish the rationale for integrating curriculum-grounded question answering, semantic retrieval, speech recognition, language generation, and speech synthesis within a single educational interaction workflow.

2.2 Domain and Theoretical Foundations

2.2.1 Artificial Intelligence in Education

Artificial intelligence in education refers to computational methods used to support teaching, learning, assessment, feedback, and educational administration. Conversational AI provides one approach through which learners can interact with intelligent systems using natural language.

However, educational AI differs from unrestricted general-purpose AI because educational correctness involves more than producing plausible answers. Responses should be appropriate to the learning objectives, students' educational levels, curriculum requirements, and learning context.

Generative language models can produce fluent explanations, but they may also generate unsupported or inaccurate information. Ji et al. [1] surveyed hallucination in natural language generation and discussed the challenges associated with generating content that is inconsistent with factual knowledge or the input context.

Consequently, the proposed system treats approved curriculum resources as an explicit source of evidence rather than relying exclusively on the internal knowledge of a generative model.

2.2.2 Large Language Models

Large Language Models (LLMs) are neural language models capable of processing and generating natural-language text. Their conversational capabilities make them suitable for interactive tutoring, explanation generation, and answering students' questions.

Recent model developments, including the Qwen2.5 family, demonstrate the continued development of general-purpose language models with different capabilities and computational requirements [2]. Such models can provide the language-generation component of an educational assistant.

Nevertheless, language-model responses are not guaranteed to be factually correct or aligned with a particular national curriculum. A model may provide an explanation that is generally valid but unsuitable for the student's grade or outside the prescribed syllabus.

This limitation motivates the integration of information retrieval with language generation so that answers can be informed by relevant curriculum content.

2.2.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with language-model generation. Rather than requiring a language model to answer solely from its learned parameters, relevant external information is retrieved and supplied as contextual evidence.

Lewis et al. [3] introduced a RAG framework that combines a pretrained sequence-to-sequence model with a dense vector index of external documents. Their work established an influential approach to incorporating retrieved knowledge into language-generation tasks.

The general RAG process consists of the following stages:

1. Receive the student's question.

2. Convert the question into a semantic representation.
3. Retrieve relevant documents or text chunks from the knowledge base.
4. Supply the retrieved context to the language model.
5. Generate a response informed by the retrieved evidence.

This approach is particularly relevant to educational applications because the retrieval collection can be restricted to approved curriculum resources.

Levonian et al. [4] investigated RAG for mathematics question answering using an external textbook. Their findings highlighted trade-offs between answer groundedness and human preference, suggesting that curriculum grounding must be balanced with clear, natural, and useful explanations.

Alawwad et al. [5] investigated textbook question answering using LLMs and RAG, demonstrating the relevance of retrieval-based approaches to questions that require information from textbook content.

Although retrieval can improve access to relevant evidence, it does not automatically guarantee that every generated response is accurate. Retrieval quality, the relevance of the selected passages, and the language model's interpretation of the retrieved information remain important considerations.

2.2.4 Semantic Embeddings

Semantic embeddings represent textual content as numerical vectors in a multidimensional space. These representations allow a system to compare texts according to their semantic similarity rather than relying exclusively on exact keyword matches.

Reimers and Gurevych [6] introduced Sentence-BERT, a sentence-embedding approach designed to generate semantically meaningful representations suitable for similarity comparison and retrieval tasks.

In the proposed system, curriculum text is divided into manageable chunks, and each chunk is transformed into an embedding. A student's question is encoded using the

same embedding model, allowing the system to identify curriculum passages that are semantically relevant to the question.

This approach is useful when students express a concept differently from the wording used in the textbook. However, semantic similarity alone does not establish that a passage contains the correct answer. The relevance of retrieved passages must therefore be evaluated in the context of the student's question.

2.2.5 Vector Similarity Search

A vector index enables efficient identification of relevant embeddings from a collection of documents. Johnson et al. [7] examined large-scale similarity search and described techniques for efficiently searching high-dimensional vector representations.

The proposed system uses FAISS to perform similarity search over curriculum embeddings. The selected configuration uses an inner-product index with L2-normalised vectors. Under this configuration, the inner product between two vectors is equivalent to their cosine similarity.

The index allows the system to retrieve candidate textbook chunks before generating an answer. Nevertheless, the highest similarity score does not necessarily indicate factual relevance, so retrieval performance must be assessed using appropriate questions and expected supporting passages.

2.2.6 Automatic Speech Recognition

Automatic Speech Recognition (ASR) converts spoken language into text so that the resulting transcription can be processed by downstream natural-language systems.

Radford et al. [8] introduced Whisper, a multilingual speech-recognition system trained using large-scale weak supervision. Their research demonstrated the potential of large-scale training to support speech recognition across multiple languages and recording conditions.

Whisper is relevant to the proposed Voice Tutor because it provides a foundation for converting spoken student questions into text. However, recognition performance can

vary depending on language, accent, pronunciation, background noise, and recording quality.

The proposed implementation uses Faster-Whisper, an implementation of Whisper based on CTranslate2 that supports optimised inference [9]. This implementation is intended to make local speech recognition more practical, although actual performance depends on the selected model, available hardware, and audio characteristics.

2.2.7 Sinhala Speech Recognition

Sinhala presents additional challenges because speech-recognition resources are more limited than those available for many high-resource languages. The performance of a multilingual model on Sinhala therefore cannot be assumed to match its performance on more extensively represented languages.

Harischandra et al. [10] investigated pretrained multilingual models for improving low-resource Sinhala ASR. Their work is relevant because the Voice Tutor is intended to accommodate Sinhala speech within a curriculum-based educational setting.

The existence of research on Sinhala ASR establishes the importance of evaluating language-specific performance rather than assuming that a multilingual speech model will provide uniformly accurate transcriptions.

Accordingly, the proposed research treats Sinhala speech recognition as an empirical evaluation problem. Its effectiveness should be assessed using representative Sinhala audio samples and appropriate transcription-quality measures.

2.2.8 Text-to-Speech

Text-to-Speech (TTS) converts textual responses into spoken output. In a voice-based educational assistant, speech synthesis completes the interaction cycle by allowing students to hear generated explanations rather than relying entirely on written responses.

OpenAI provides a text-to-speech API that can generate spoken audio from text input [11]. Browser-based speech synthesis provides an alternative mechanism through the

Web Speech API, although voice availability and quality depend on the browser, operating system, and installed speech services.

The proposed system incorporates a cloud-based TTS option and a browser speech-synthesis fallback to improve the availability of spoken responses. The selected output method must also be evaluated for pronunciation, intelligibility, language support, response latency, and operational cost.

2.3 Related research

2.3.1 Curriculum-Grounded Educational Question Answering

Curriculum-grounded question answering aims to generate responses using information from educational resources such as textbooks and teacher guides. This approach is particularly important when students require explanations that correspond to a prescribed syllabus.

Levonian et al. [4] investigated RAG for middle-school mathematics using an external textbook. Their work highlighted the importance of balancing answer groundedness with human preferences when designing educational question-answering systems.

Alawwad et al. [5] examined textbook question answering using LLMs and RAG. Their work supports the use of retrieval-based techniques for educational questions that require information distributed across textbook content.

The broader RAG framework introduced by Lewis et al. [3] provides the technical foundation for combining retrieved evidence with language generation. In addition, Ji et al. [1] discussed hallucination as a persistent challenge in natural-language generation, reinforcing the need to evaluate the factual support of generated answers.

For the proposed system, the National Institute of Education's Grade 10 and Grade 11 Science textbooks and corresponding teacher guides provide the intended curriculum knowledge sources [12]–[15]. These resources establish the subject-specific scope of the knowledge base and provide material against which generated explanations can be checked.

The proposed system therefore aims to combine semantic retrieval with curriculum-informed generation rather than relying exclusively on a general-purpose chatbot. Its effectiveness depends on the completeness of the indexed resources, the quality of retrieval, and the correctness of the resulting explanations.

2.3.2 Multilingual and Sinhala Automatic Speech Recognition

Multilingual ASR enables users to interact with systems through spoken language without requiring manual transcription. However, recognition quality can differ substantially across languages and acoustic conditions.

Harischandra et al. [10] investigated pretrained multilingual models for low-resource Sinhala speech recognition and enhancement. Their research is directly relevant to the Sinhala voice-input requirements of the proposed system.

Whisper provides a broader multilingual ASR foundation [8], while Faster-Whisper offers an implementation intended to improve inference efficiency [9]. These technologies provide practical options for integrating speech input into the Voice Tutor.

However, the availability of multilingual speech-recognition models does not guarantee reliable Sinhala transcription for every user or recording environment. Sinhala ASR should therefore be evaluated independently using representative speech samples, with attention to transcription errors and their effects on subsequent question answering.

2.3.3 Voice-Based Educational Assistants

Voice-based educational assistants combine speech recognition, natural-language processing, response generation, and speech synthesis to enable spoken interaction between learners and educational systems.

The technologies reviewed in this chapter provide the main building blocks for such systems. Whisper supports speech recognition [8], Faster-Whisper provides an implementation for local transcription [9], and text-to-speech services support spoken responses [11].

The proposed research places particular emphasis on combining these interaction capabilities with curriculum-grounded question answering. Rather than treating speech interaction as a standalone feature, the system integrates voice input with retrieval from approved Sri Lankan Science curriculum resources.

This integration is intended to allow students to ask questions conversationally and receive explanations informed by relevant textbook content. The resulting system must be evaluated as a complete workflow because speech-recognition errors can affect retrieval, and retrieval errors can affect the quality of the generated explanation.

2.3.4 Offline and Resource-Conscious AI

Offline educational assistants are relevant in environments where internet connectivity, privacy, or access to external AI services may be constrained. Local inference can reduce dependence on remote services, although it may introduce limitations related to model size, memory usage, processing speed, and response quality.

Ollama provides infrastructure for running supported language models locally [16]. This capability is relevant to the proposed architecture because it enables local language-model inference when a compatible model and sufficient computational resources are available.

The proposed system follows a hybrid strategy in which local components are prioritised where practical and external services are used where appropriate. For example, local speech recognition and local language-model inference can reduce dependence on cloud services, whereas cloud-based speech synthesis can provide an alternative when local speech output is unsuitable.

The trade-offs between local and cloud processing include computational cost, internet dependency, privacy, response latency, and model capability. These considerations must be assessed against the system's intended educational setting.

2.4 Existing systems products and techniques

Several established technologies provide the foundation for the proposed system.

Table 1 - Existing systems products and techniques

Technology or approach	Main capability	Relevance to the Voice Tutor	Limitation or consideration
General-purpose LLM chatbots	Natural language question answering	Provides a conversational interaction model	May generate unsupported or out-of-syllabus information
Retrieval-Augmented Generation	Grounded response generation using external documents	Connects answers with official curriculum content	Depends on retrieval quality and correct context interpretation
OpenAI Whisper	Multilingual speech recognition	Provides the basis for transcription	Accuracy varies by language and recording conditions
faster-whisper	Optimized Whisper inference	Supports local CPU-based speech recognition	Still requires sufficient processing resources
FAISS	Efficient vector similarity search	Retrieves relevant textbook chunks	Similarity does not necessarily indicate factual relevance
Ollama	Local language model serving	Supports offline or local reasoning	Model quality and speed depend on hardware and model choice
OpenAI text-to-speech	Cloud speech synthesis	Produces spoken educational responses	Requires connectivity and API access
Web Speech	Browser-native	Provides an alternative speech	Voice availability and quality vary across

API	speech synthesis	output mechanism	devices

These technologies are not direct competitors in every respect. Some are underlying tools, while others represent broader system architectures. Their relevance is assessed according to their role in the proposed educational voice interaction workflow.

2.5 Gap synthesis and conceptual framework

The reviewed literature identifies several requirements for the development of a curriculum-grounded Voice Tutor.

First, educational responses require reliable grounding in authoritative educational resources. RAG provides a mechanism for supplying relevant external evidence to a language model [3], [4], [5]. However, the system must still evaluate whether the retrieved passages support the generated explanation.

Second, natural-language generation can support conversational educational interaction, but hallucination remains a significant limitation [1]. This motivates the use of curriculum evidence, careful response generation, and evaluation of answer correctness.

Third, Sinhala speech processing requires explicit evaluation. Research into pretrained multilingual models demonstrates the relevance of transfer learning for low-resource Sinhala ASR [10], but the proposed system must establish its own transcription performance under its intended audio conditions.

Fourth, semantic embeddings and vector similarity search provide a technical basis for retrieving relevant textbook content [6], [7]. Nevertheless, retrieval effectiveness depends on text extraction, chunking, embedding quality, index configuration, and the relationship between the student's question and the available curriculum material.

Fifth, voice-based interaction requires the integration of speech recognition, question answering, and speech synthesis. Whisper and Faster-Whisper provide speech-recognition options [8], [9], while cloud-based TTS provides a mechanism for generating spoken responses [11].

Finally, the choice between local and cloud-based AI services involves trade-offs among privacy, connectivity, computational resources, latency, and response quality. Local language-model serving through Ollama provides one option for reducing reliance on external inference services [16].

Based on these findings, the proposed Voice Tutor adopts a curriculum-grounded conversational architecture that integrates Sinhala speech input, semantic retrieval from approved educational resources, language-model-based explanation generation, and speech output. The architecture is intended to make educational interaction more accessible while maintaining a clear relationship between generated explanations and the prescribed curriculum.

The research contribution is therefore not the introduction of an entirely new ASR, embedding, vector-search, or language-generation algorithm. Instead, it lies in integrating and evaluating these technologies within a curriculum-focused educational voice-interaction system designed around Sri Lankan Grade 10 and Grade 11 Science resources.

3 RESEARCH METHODOLOGY

3.1 Research design

This research follows an applied design and development approach supported by experimental evaluation.

The applied research component focuses on addressing a practical educational problem through the development of a functional software system. The experimental component evaluates alternative local language models and assesses technical performance where suitable measurements are available.

The research is organized into five phases.

Phase 1: Requirements analysis and literature review

Identify educational requirements, review relevant AI techniques, examine the available curriculum resources, and establish the technical requirements of the Voice Tutor.

Phase 2: System architecture and component design

Design the frontend and backend interaction workflow, document-processing pipeline, semantic retrieval mechanism, language model integration, multilingual response behavior, and speech synthesis architecture.

Phase 3: Implementation and integration

Develop and integrate the speech-processing, retrieval, conversational reasoning, API, and speech synthesis subsystems.

Phase 4: Model comparison and system validation

Compare the three selected local language models using common STEM prompts. Perform functional testing and collect system performance measurements where available.

Phase 5: Analysis and documentation

Analyze the model comparison results, evaluate the technical implementation against the research objectives, document limitations, and identify future improvements.

This research design is suitable for evaluating both the practical feasibility of the proposed architecture and the relative performance of alternative local language models.

3.2 Data participants or study materials

3.2.1 Curriculum materials

The primary knowledge sources consist of Sri Lankan National Institute of Education Science resources, including:

- Grade 10 Science textbooks;
- Grade 11 Science textbooks;
- Grade 10 Science Teacher's Guide; and
- Grade 11 Science Teacher's Guide.

These resources form the authoritative retrieval collection for curriculum-grounded responses.

3.2.2 Audio data

The evaluation should use a controlled collection of English and Sinhala Science questions relevant to the Grade 10–11 curriculum.

Where original participant recordings are collected, participant consent and appropriate institutional procedures must be followed.

The final thesis should report:

- number of recordings;
- language distribution;
- recording conditions;
- speaker characteristics at an appropriate non-identifying level;
- question categories;
- duration statistics; and
- data-splitting procedure.

No such quantities should be invented before the actual evaluation is completed.

3.2.3 Evaluation questions

Evaluation questions should cover:

- factual questions;
- conceptual questions;
- definition questions;
- comparison questions;
- explanation questions;
- multi-step questions;
- bilingual questions; and
- out-of-syllabus questions.

3.3 Experimental or study procedure

The evaluation procedure consists of the following stages.

Stage 1: Audio input

A participant or test case provides a spoken Science question.

Stage 2: Speech recognition

The audio is processed by the Faster-Whisper pipeline.

Stage 3: Retrieval

The recognised question is embedded and compared with the curriculum embedding index.

The five highest-ranking relevant chunks are retrieved.

Stage 4: Response generation

The retrieved chunks and the user question are passed to the selected conversational model.

The model is instructed to:

- behave as an O/L Science tutor;
- use the retrieved curriculum evidence;
- avoid unsupported information;
- answer at an appropriate educational level;
- respect the requested language; and
- refuse questions for which sufficient curriculum evidence is unavailable.

Stage 5: Speech synthesis

The generated answer is passed to the TTS layer.

Stage 6: Evaluation

The output is assessed using predefined technical and qualitative measures.

3.4 Measures and evaluation metrics

3.4.1 Speech recognition

The principal speech-recognition metric is Word Error Rate:

WER = (S + D + I) / N

where:

- S = substitutions;
- D = deletions;
- I = insertions; and
- N = number of words in the reference transcription.

Character Error Rate may also be reported for Sinhala where appropriate:

CER = (S + D + I) / N

The final measured values should be inserted after testing.

3.4.2 Retrieval evaluation

Retrieval can be evaluated using:

- Recall@K;

- Precision@K;
- Mean Reciprocal Rank;
- relevance judgement; and
- retrieval similarity score.

If a manually labelled evaluation set is created, each question should be associated with one or more expected relevant curriculum passages.

3.4.3 Answer quality

Generated answers can be evaluated using criteria such as:

- factual correctness;
- curriculum relevance;
- completeness;
- clarity;
- language appropriateness;
- grounding in retrieved evidence; and
- unsupported claims.

A structured rubric should be defined before human evaluation.

3.4.4 System performance

Technical measurements may include:

- speech-recognition processing time;
- retrieval time;
- language-model response time;
- TTS generation time;
- end-to-end response time; and
- memory/CPU utilisation where measurable.

3.4.5 Functional correctness

Functional testing verifies whether:

- valid audio can be transcribed;
- invalid audio is rejected appropriately;
- questions are retrieved correctly;
- answers are generated;
- language selection operates correctly;
- TTS functions;
- fallback behaviour operates;
- out-of-syllabus questions are handled appropriately; and
- API responses follow their defined schemas.

3.5 Data analysis

The model comparison is analyzed through descriptive comparison of the reported evaluation scores.

The analysis considers the trade-off between response quality and operational efficiency rather than focusing exclusively on one metric.

The overall scores in the supplied report are used to identify the model selected for the local deployment configuration.

For the broader Voice Tutor evaluation, quantitative measures such as WER, retrieval Precision@k, latency, and resource usage should be summarized using appropriate descriptive statistics.

Where repeated trials are available, the analysis can include mean, median, standard deviation, and confidence intervals, depending on the data distribution and sample size.

Qualitative assessment may be used to examine response clarity, explanation structure, Sinhala language quality, and the appropriateness of refusal responses.

The final analysis should distinguish measured observations from design expectations and interpretations.

3.6 Ethics privacy security and responsible AI

The Voice Tutor is intended for school students, so privacy, security, and responsible AI considerations are important.

3.6.1 Student audio privacy

The local transcription architecture is designed to process uploaded audio using faster-whisper on the backend without sending the raw recording to an external transcription provider.

The system description states that recordings are not stored permanently. This behavior should be confirmed by reviewing the actual upload handling, temporary file cleanup, server logs, and deployment storage configuration.

If cloud-based language generation or speech synthesis is enabled, the system may transmit the student's text question or generated response to an external provider. This distinction should be communicated clearly to users.

3.6.2 Data minimization

The system should collect only the information necessary to provide tutoring services.

Student names, contact information, and other identifying details should not be required for ordinary question answering.

Where logs are retained for debugging or evaluation, they should avoid including unnecessary personal information and should follow an appropriate retention policy.

3.6.3 Curriculum-grounded response generation

The retrieval system is intended to reduce unsupported answers by supplying official curriculum context.

The system prompt instructs the model to avoid inventing outside facts and to use a predefined refusal response when the requested topic is not covered by the available material.

However, prompt-level guardrails are not a complete safety mechanism. The system should also be evaluated for unsupported answers, misleading explanations, incorrect refusals, and failures to identify missing context.

3.6.4 Language accessibility

Sinhala and English response modes are included to support the linguistic requirements of the intended user group.

The quality of Sinhala responses should be assessed by speakers familiar with the target educational context. Automatic language selection should also be tested with mixed-language and code-switched questions.

3.6.5 Student protection

As the target users are minors, any formal student testing should obtain the approvals and permissions required by the university and relevant educational authorities.

The research should use age-appropriate test procedures and ensure that participation is voluntary, appropriately supervised, and separate from academic assessment.

3.7 Reproducibility and research artefacts

Reproducibility is supported by documenting the software environment, model configurations, knowledge sources, evaluation prompts, and scoring procedures.

The following artefacts are relevant to the study:

Table 2 - Reproducibility and research artefacts

Artefact	**Purpose**
Curriculum PDF collection	Records the knowledge sources
Document extraction scripts	Reproduce the text preparation process
Chunking configuration	Defines chunk size and overlap
Embedding model configuration	Identifies the vector representation method
Index manifest	Records index compatibility information

FAISS index	Stores the searchable document vectors
Model comparison prompt set	Ensures consistent model testing
Scoring rubric	Defines evaluation criteria
Test cases	Supports functional verification
Environment and dependency records	Supports deployment reproducibility
Evaluation logs	Preserve measured results and observations

The final thesis should include actual software versions, model identifiers, hardware specifications, and test configuration details rather than relying only on approximate version ranges.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The Voice Tutor requirements are divided into functional and non-functional requirements.

4.1.1 Functional requirements

Table 3 - Functional requirements

ID	Requirement	Description	Priority
FR01	Audio capture	Capture student speech through the browser microphone	High
FR02	Audio upload	Submit recorded audio to the backend	High
FR03	Audio validation	Validate supported formats and file sizes	High
FR04	Speech transcription	Convert uploaded audio into text	High
FR05	VAD fallback	Retry transcription without VAD when the filtered result is empty	Medium
FR06	Curriculum retrieval	Retrieve relevant Science textbook chunks	High
FR07	Conversational reasoning	Generate responses using retrieved context	High
FR08	Language selection	Support English, Sinhala, and automatic response modes	High
FR09	Refusal handling	Provide a predefined response when the requested topic cannot be answered from available resources	High
FR10	Source metadata	Return available source information with the answer	Medium

FR11	Text-to-speech	Convert generated responses into spoken audio	High
FR12	Speech fallback	Use browser-native speech synthesis when cloud TTS is unavailable	Medium
FR13	Local inference	Support locally deployed language models	High
FR14	Cloud inference	Support the configured cloud language model when enabled	Medium
FR15	Health monitoring	Report relevant backend and knowledge-index status	Medium

4.1.2 Non-functional requirements

Performance: The system should provide responses within an acceptable timeframe for interactive educational use.

Reliability: The architecture should handle unavailable external services and invalid user inputs without terminating the complete interaction.

Usability: The interface should allow students to record questions and understand responses without extensive technical knowledge.

Maintainability: Individual services should be separated to support independent testing and modification.

Scalability: The backend should allow future improvements in model serving, retrieval, and concurrent request processing.

Privacy: Local speech recognition should avoid unnecessary transmission of raw audio to external services.

Portability: The browser interface should work on supported desktop and mobile browsers, subject to microphone and speech API availability.

Curriculum alignment: Responses should be grounded in the selected official learning materials and communicate uncertainty or unsupported topics appropriately.

4.2 Component architecture within the complete solution

The Voice Tutor operates as an individual component of the Study_Buddy ecosystem.

Its main interfaces are the browser-based frontend and the FastAPI backend.

The frontend handles microphone access, recording, API requests, response presentation, and speech playback.

The backend coordinates audio validation, speech recognition, curriculum retrieval, conversational reasoning, and speech synthesis.

The system can be represented through the following architecture.

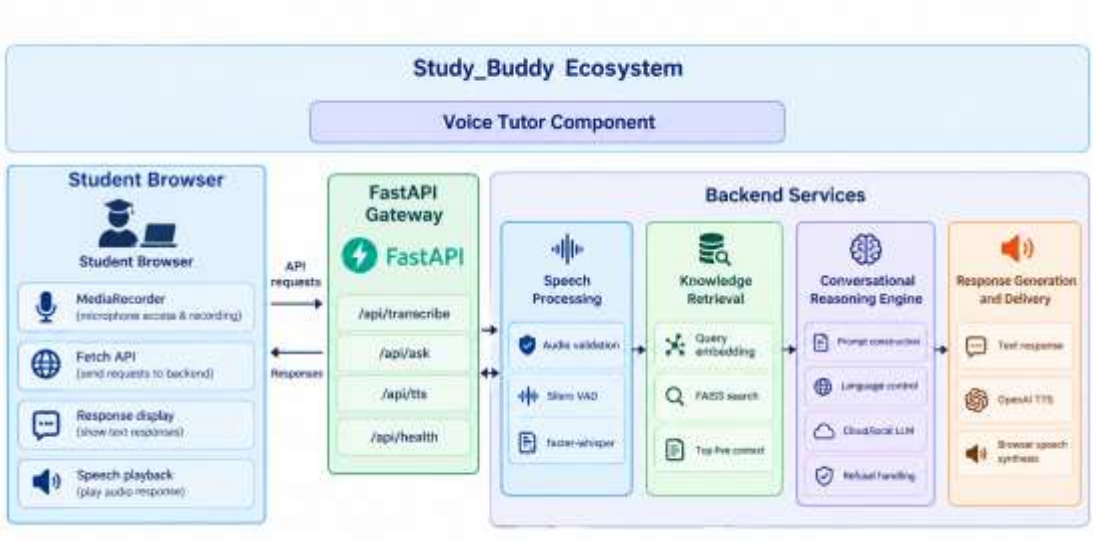


Figure 3 - component Architecture

The architecture separates computational responsibilities into distinct services. This supports independent testing and provides flexibility when replacing models or modifying individual processing stages.

4.3 Design alternatives and decisions

4.3.1 Cloud-based versus local speech recognition

Cloud speech recognition can provide managed infrastructure and may offer strong recognition performance for supported languages. However, it requires network connectivity and may introduce recurring costs and external audio processing.

Local speech recognition provides greater control over the audio processing pipeline and avoids per-request transcription charges.

The proposed system selects faster-whisper with CPU-based INT8 inference as its primary transcription mechanism.

This decision is based on the requirements for reduced cloud dependency, privacy-oriented processing, and local deployment rather than a claim that local transcription is always faster or more accurate than cloud alternatives.

4.3.2 Keyword retrieval versus dense retrieval

Keyword-based retrieval methods identify documents using lexical matching. They can be effective when queries share vocabulary with the source documents.

Dense semantic retrieval uses vector representations to identify related passages even when wording differs.

The proposed system uses dense retrieval with all-MiniLM-L6-v2 and FAISS to support natural-language student questions.

A hybrid approach combining lexical and dense retrieval could be investigated in future work, especially for scientific terms and Sinhala-English code-switching.

4.3.3 Cloud LLM versus local LLM

Cloud language models can offer strong language generation capabilities without requiring local GPU infrastructure. However, they introduce external service dependency, usage costs, and potential data-transfer considerations.

Local language models allow inference to occur within the deployment environment and can support offline operation.

The proposed architecture supports both options, with the local model comparison identifying Qwen2.5 7B as the preferred candidate according to the supplied benchmark.

4.3.4 Language model selection

The three local models were compared using five evaluation criteria.

Table 4 - Language model selection

Model	Correctness	Relevance	Explanation	Response time	Efficiency	Overall
Mistral 7B	92	91	93	65	72	82.6
Qwen2.5 7B	89	90	90	88	86	**88.6**
Llama 3.2 3B	78	76	74	95	94	83.4

Consequently, Qwen2.5 7B was selected as the preferred local model for the proposed tutoring architecture.

4.3.5 Cloud versus browser-native speech synthesis

Cloud synthesis can provide a consistent managed speech-generation service where connectivity and credentials are available.

Browser-native synthesis offers a local alternative that avoids an external synthesis request. However, its voice selection and language coverage vary across devices.

The proposed system combines both approaches. OpenAI tts-1 is used as the cloud synthesis option, while browser-native synthesis provides a fallback where supported.

4.4 Detailed design

4.4.1 Audio capture

The browser uses the MediaRecorder API to capture student speech.

WebM/Opus is preferred because it provides practical browser compatibility and compressed audio size.

The recorded audio is packaged using FormData and submitted to:

POST /api/transcribe

4.4.2 Audio validation

The backend verifies:

- file presence;
- file extension;
- supported media type;
- file size; and
- successful decoding.

A maximum file size of 25 MB is applied.

Supported formats include:

- WebM
- WAV
- MP3
- OGG
- M4A
- FLAC
- MP4
- AAC

4.4.3 Speech recognition

Faster-Whisper is configured using:

- base model
- CPU execution
- INT8 computation
- Silero VAD
- beam size of 5

A VAD configuration is used to remove non-speech portions of the recording.

If a non-empty recording unexpectedly produces empty transcription, the system retries without VAD filtering.

This fallback improves robustness against recordings where speech segmentation is unsuccessful.

4.4.4 Curriculum knowledge-base construction

The curriculum knowledge base is constructed from:

- Grade 10 Science textbook
- Grade 11 Science textbook
- Grade 10 Science Teacher's Guide
- Grade 11 Science Teacher's Guide

PDF content is extracted using PyMuPDF.

The extracted content is divided into approximately 200–400-word chunks.

Each chunk is associated with metadata including:

- chunk ID
- grade
- subject area
- topic
- subtopic
- document type
- source file

- starting page
- ending page

4.4.5 Embedding generation

The primary local embedding model is:

sentence-transformers/all-MiniLM-L6-v2

The model generates 384-dimensional embeddings.

An alternative cloud configuration using text-embedding-3-small can produce 1536-dimensional embeddings.

The selected query encoder must match the encoder used to construct the FAISS index.

4.4.6 FAISS retrieval

The system uses: IndexFlatIP

The vectors are L2 normalised before insertion.

This makes inner-product similarity equivalent to cosine similarity.

For each query, the system retrieves the top five candidate chunks.

4.4.7 Retrieval-Augmented Generation

The retrieved chunks are combined with the student's question.

The language model receives instructions to act as a Sri Lankan O/L Science tutor.

The prompt instructs the model to:

- priorities retrieved curriculum content;
- avoid unsupported claims;
- provide understandable explanations;
- answer at an appropriate educational level;
- use the requested language; and
- refuse questions that cannot be supported by the retrieved syllabus resources.

4.4.8 Out-of-syllabus handling

A major safety mechanism is controlled refusal.

When relevant evidence is unavailable, the system should not generate an unrestricted answer.

The English refusal is:

“This is not in the syllabus.”

The Sinhala refusal is:

“මෙය විෂය මාලාවේ හෝ ලබා දී ඇති සම්පත්වල අන්තගර්ත නොවේ.”

This behavior reduces the risk of presenting general model knowledge as curriculum-authorized information.

4.4.9 Multilingual processing

The system supports:

- English;
- Sinhala; and
- automatic language selection.

The request model uses:

auto, en, or si.

The language selected for the response is passed through the generation and TTS layers.

4.4.10 Text-to-speech

The primary TTS implementation uses a cloud speech service.

A browser Web Speech API fallback is provided when the cloud TTS service is unavailable.

For Sinhala responses, the system attempts to use a Sinhala-compatible browser voice where available.

4.4.11 API design

The major endpoints are:

Endpoint	Method	Purpose
/api/transcribe	POST	Convert uploaded audio into text
/api/ask	POST	Process question and generate answer
/api/tts	POST	Convert text into speech
/api/health	GET	Check backend/index health

The /api/ask endpoint accepts:

- question;
- response_language.

The response contains:

- answer;
- sources;
- images;
- visual_required; and
- visual.

4.5 Standards and professional practice

The implementation follows modular software engineering principles.

The system separates:

- presentation;
- API handling;
- speech processing;
- retrieval;
- language generation; and
- TTS.

This separation improves maintainability and allows individual services to be tested independently.

The research also follows professional practices including:

- source acknowledgement;
- secure handling of credentials;
- separation of configuration from source code;
- API validation;
- error handling;
- reproducibility;
- controlled logging; and
- responsible AI design.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 5 - technology stack

Layer	Technology
Frontend	HTML5, CSS3, JavaScript
Audio capture	MediaRecorder API
Backend	Python, FastAPI
Server	Uvicorn
ASR	Faster-Whisper
VAD	Silero VAD
PDF processing	PyMuPDF
Embeddings	Sentence Transformers
Vector search	FAISS
LLM	OpenAI / local Ollama model
Local inference	Ollama
TTS	Cloud TTS / Web Speech API
Data format	JSON / JSONL
API format	REST/JSON

5.2 Component implementation

5.2.1 Frontend audio capture

The frontend uses the browser's microphone permission system to capture student speech.

The recording process is designed to:

1. Request microphone access.
2. Initialize MediaRecorder.
3. Select a supported recording MIME type.
4. Collect audio data in chunks.
5. Stop recording when the student completes the question.
6. Combine the collected data into a Blob.
7. Submit the Blob to the transcription endpoint.

The format selection logic prioritizes WebM with Opus when the browser supports it.

This approach is intended to reduce unnecessary audio-processing complexity and provide compatibility across commonly used browsers.

5.2.2 Audio upload and validation

The backend accepts uploaded recordings through the transcription endpoint.

Validation is applied before the audio is passed to the speech recognition engine.

The main checks include:

- Non-empty file content.
- Supported file extension.
- Maximum payload size.
- Successful audio preparation for inference.

Invalid input is rejected through a structured error response rather than being passed directly to the model.

5.2.3 Speech transcription

The transcription service loads the configured faster-whisper model and performs inference using CPU-based INT8 computation.

Silero VAD is enabled to identify speech segments and remove unnecessary silence from the transcription process.

If the initial VAD-filtered pass returns an empty transcript, the system retries the transcription without VAD filtering.

The fallback is intended to recover quiet speech that may have been excluded during speech activity detection.

5.2.4 Curriculum document processing

The document processing implementation is separated into the PDF extraction and chunking stages.

The extraction service loads the selected learning resources and retrieves their text.

The chunking service divides the extracted text into manageable sections while preserving source references.

Metadata is associated with each chunk to support curriculum categorization and source reporting.

The resulting records are passed to the embedding generation subsystem.

5.2.5 Embedding and vector index implementation

The embedding service generates dense vectors using the selected sentence-transformer model.

The vector store uses FAISS IndexFlatIP to support inner-product search.

L2 normalization is applied so that the inner product corresponds to cosine similarity.

The index manifest records the embedding model and vector dimension.

This design allows the system to identify mismatches between the indexing and query-embedding configurations.

5.2.6 Language model integration

The language model service supports cloud and local inference options.

The cloud configuration uses OpenAI gpt-4o-mini with a low temperature setting of 0.1.

The local configuration communicates with an Ollama service hosted at:

http://127.0.0.1:11434/api/chat

The local model comparison identifies Qwen2.5 7B as the selected model based on the reported overall score.

The system prompt defines the tutor's role, curriculum constraints, language requirements, and refusal behavior.

The implementation should ensure that the active model configuration is clearly identifiable during testing, since the cloud model and the local models may produce different response characteristics.

5.2.7 API endpoint implementation

The backend exposes four main endpoints.

Transcription endpoint

POST /api/transcribe

Accepts multipart audio uploads and returns a structured transcription response.

Question-answering endpoint

POST /api/ask

Accepts a question and response-language setting. It invokes retrieval and language generation and returns the answer with available source metadata.

Text-to-speech endpoint

POST /api/tts

Accepts text and a voice configuration. When the cloud synthesis service is available, the endpoint returns generated audio in an audio/mpeg format.

Health endpoint

GET /api/health

Reports system status and whether the knowledge index has been loaded. Additional checks may be required to confirm language model availability and service readiness.

5.3 Integration with the other components

The Voice Tutor forms part of the broader Study_Buddy ecosystem.

Its integration responsibilities include accepting student questions, retrieving relevant curriculum knowledge, returning structured answers, and supporting any relevant visual explanation features exposed through the wider application.

The API response model includes optional image and visual fields:

- images
- visual_required
- visual

These fields provide a potential interface for integration with other Study_Buddy capabilities.

The Voice Tutor's main responsibility remains conversational interaction. The availability and behavior of visual explanation features depend on the implementation of the other ecosystem components.

Integration should be verified through the actual exchange of API requests and responses rather than inferred from the presence of compatible data structures.

5.4 Verification and software testing

The testing strategy should cover individual services, API behavior, integration, and complete user interactions.

5.4.1 Unit testing

Unit tests should focus on individual functions and processing modules.

Examples include:

- Audio file validation.
- Empty payload detection.
- Extension and file-size restrictions.
- Chunk creation and metadata preservation.
- Embedding vector dimensionality checks.
- Vector normalization.
- Language-mode validation.
- Refusal response selection.
- Source metadata serialization.

5.4.2 API testing

API tests should verify request validation, response structure, status codes, and failure handling.

Table 6 - API testing

Test ID	Endpoint	Test scenario	Expected behavior
API01	/api/transcribe	Valid supported audio	Return transcription response
API02	/api/transcribe	Empty recording	Return appropriate error
API03	/api/transcribe	Unsupported extension	Reject upload
API04	/api/transcribe	Oversized upload	Reject request
API05	/api/ask	Valid English question	Return structured answer
API06	/api/ask	Sinhala response mode	Apply Sinhala response instructions
API07	/api/ask	Automatic language	Select a response language based

		mode	on the question
API08	/api/ask	Out-of-scope question	Apply the intended refusal policy
API09	/api/tts	Valid text and voice	Return audio when the service is available
API10	/api/health	Knowledge index loaded	Report the relevant status

These are proposed test cases. Their actual pass/fail outcomes must be recorded after execution.

5.4.3 Retrieval testing

Retrieval tests should examine whether the selected chunks correspond to the intended Science topics.

A labeled dataset can be prepared using questions whose relevant textbook passages are known.

Testing should cover:

- Direct factual questions.
- Conceptual questions.
- Questions requiring information from multiple passages.
- Questions with paraphrased terminology.
- Sinhala and English queries.
- Questions with scientific vocabulary and mixed-language wording.
- Questions that fall outside the indexed documents.

The evaluation should identify both successful retrievals and common retrieval failures.

5.4.4 Language model testing

The model comparison should be repeated under controlled conditions where possible.

All candidate models should receive the same prompt set and comparable contextual inputs.

The test should document:

- Model identifiers and versions.
- Prompt content.
- System instructions.
- Hardware specifications.
- Temperature and decoding settings.
- Number of repeated trials.
- Evaluation rubric.
- Timing methodology.
- Resource measurement methodology.

Response correctness and contextual grounding should be assessed independently of response fluency.

5.4.5 End-to-end testing

End-to-end testing should verify the complete voice interaction from recording to audio playback.

A representative test involves asking a question about a Grade 10 or Grade 11 Science topic, confirming the transcription, checking the retrieved context, inspecting the generated answer, and verifying speech output.

Failure cases should also be included, such as microphone permission denial, unavailable cloud credentials, local model failure, empty recordings, and missing browser speech voices.

5.5 Deployment and operation

The backend is designed to run through Uvicorn and expose the FastAPI endpoints.

The frontend communicates with the backend through asynchronous HTTP requests.

The local inference components include the faster-whisper transcription engine, the embedding model, the FAISS index, and an Ollama-hosted language model where the local configuration is active.

The cloud configuration requires valid API credentials and network connectivity.

A deployment should ensure that:

- The required Python dependencies are installed.
- The selected speech recognition and embedding models are available.
- The knowledge index is generated and loaded.
- The local language model is available when local inference is selected.
- API credentials are stored securely.
- The frontend can reach the backend.
- Microphone access is enabled in the browser.
- Upload limits and service timeouts are configured appropriately.

The current research description establishes the intended deployment architecture but does not provide sufficient information to claim that the component has been deployed and load-tested in a production school environment.

6 RESULTS AND VALIDATION

6.1 Research results

6.1.1 Local language model comparison

The supplied local LLM benchmark report evaluates three models using the same STEM educational prompts.

The reported scores are summarized below.

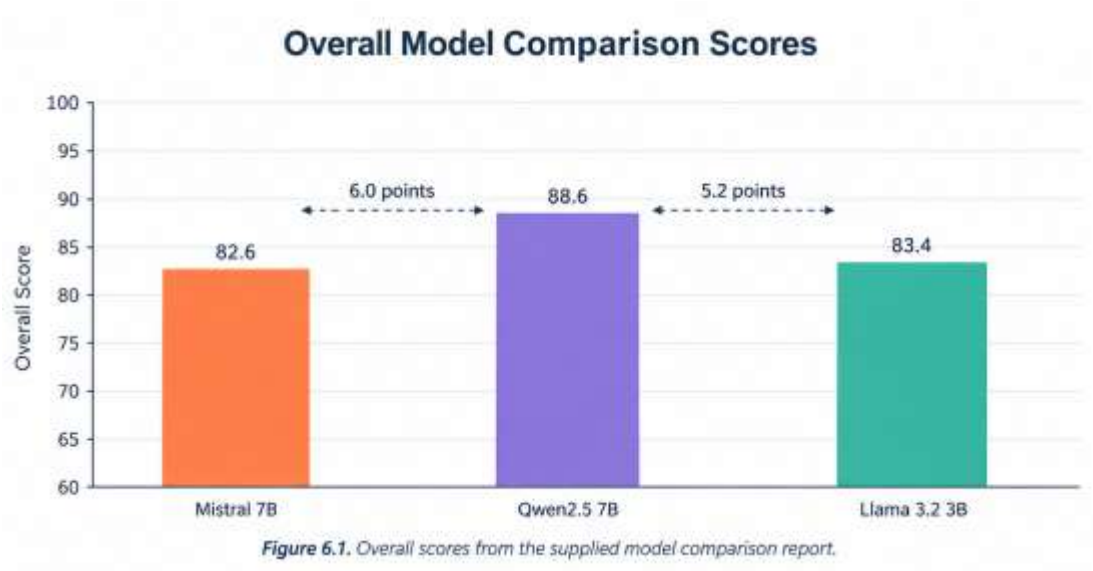


Figure 4 - Local language model comparison

Overall model comparison scores

Reported scores from the supplied local LLM benchmark evaluation.

0255075100Mistral 7BQwen2.5 7BLlama 3.2 3B

Figure 6.1. Overall scores from the supplied model comparison report.

Qwen2.5 7B achieved the highest reported overall score of 88.6. Mistral 7B achieved 82.6, while Llama 3.2 3B achieved 83.4.

The reported difference between Qwen2.5 7B and Mistral 7B is 6.0 points. The difference between Qwen2.5 7B and Llama 3.2 3B is 5.2 points.

The results indicate that Qwen2.5 7B achieved the most favorable aggregate score under the evaluation approach documented in the comparison report.

6.1.2 Criterion-level comparison

Model performance by evaluation criterion

Scores reported in the supplied model comparison document.

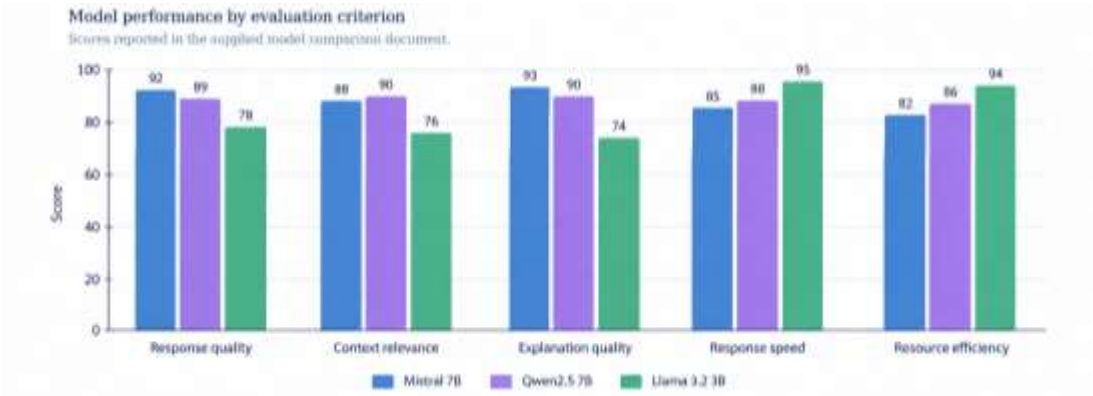


Figure 5 - Criterion-level model comparison

Mistral 7B achieved the highest scores for response quality and explanation quality. Its response quality score was 92, while its explanation quality score was 93.

Qwen2.5 7B received scores of 89 for response quality, 90 for contextual relevance, and 90 for explanation quality. Its speed score of 88 and resource-efficiency score of 86 indicate a more favorable operational profile than the reported Mistral scores.

Llama 3.2 3B achieved the highest scores for response speed and resource efficiency, receiving 95 and 94, respectively. However, its reported response quality, contextual relevance, and explanation quality scores were lower than those of the other models.

The results illustrate the trade-off between educational response characteristics and computational efficiency.

6.2 Product and technical results

The implemented system provides:

- browser-based voice capture
- automatic transcription
- semantic curriculum retrieval
- curriculum-grounded answer generation
- English/Sinhala response handling

- text-to-speech output
- local/cloud fallback mechanisms
- out-of-syllabus refusal
- API-based integration

6.2.1 Model comparison

The development-stage model comparison identified Qwen2.5 7B as the highest-scoring candidate under the defined evaluation criteria.

The overall score was 88.6 compared with 83.4 for Llama 3.2 3B and 82.6 for Mistral 7B.

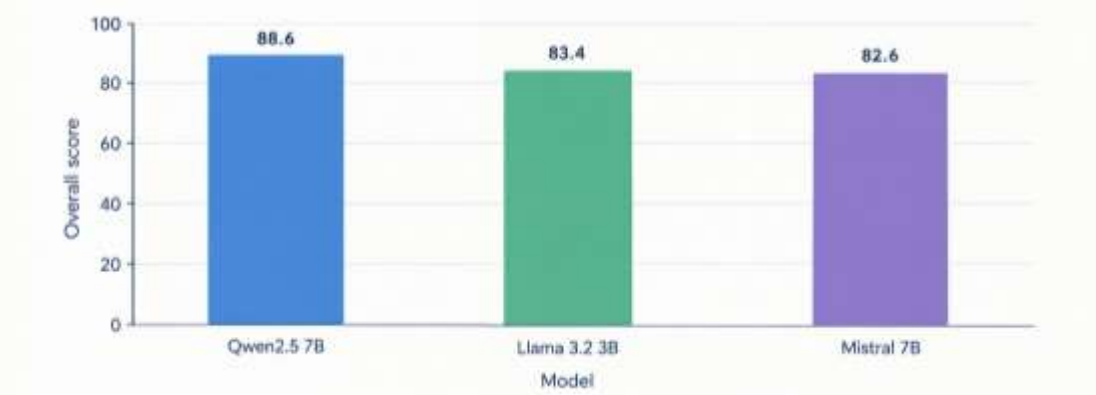


Figure 6 - Model comparison

6.3 User or stakeholder validation

6.4 Integration results

The proposed architecture connects the frontend, backend, speech recognition, retrieval, reasoning, and speech synthesis subsystems.

The main integration workflow is based on the following sequence:

- Browser audio capture and upload.
- Local transcription.
- Question submission.
- Curriculum retrieval.
- Contextual answer generation.
- Structured response delivery.

- Speech synthesis and playback.

The backend API contracts provide a common communication structure for these stages.

However, the supplied material does not include integration logs, execution screenshots, test summaries, or measured success rates. Full integration success should therefore be confirmed using end-to-end test evidence before this section is finalized.

6.5 Summary of findings

The available model comparison results identify Qwen2.5 7B as the preferred local model under the reported evaluation approach.

The comparison demonstrates a clear trade-off among the three candidates. Mistral 7B achieved the strongest reported response-quality and explanation-quality scores, Llama 3.2 3B achieved the highest reported speed and efficiency scores, and Qwen2.5 7B achieved the highest overall score.

The technical design establishes a coherent architecture for integrating local speech recognition, curriculum-based semantic retrieval, constrained language generation, multilingual response modes, and alternative speech synthesis mechanisms.

7 DISCUSSION

7.1 Interpretation of findings

The principal contribution of the research is the integration of several technologies into a curriculum-oriented conversational voice tutoring architecture.

The system does not rely exclusively on the language model's internal knowledge. Instead, curriculum content is explicitly retrieved before generation.

This architectural decision is significant because educational correctness requires contextual alignment with prescribed learning materials.

The use of semantic retrieval also reduces dependence on exact keyword matching. A student may formulate a question differently from the wording used in a textbook while still referring to the same underlying concept.

The voice interface extends this approach by allowing the learner to interact through spoken language.

However, the quality of the complete system remains dependent on the quality of each stage. A transcription error can affect retrieval; an incorrect retrieval can affect generation; and a correct generated answer may still be unsuitable if the language or explanation level is inappropriate.

Therefore, the Voice Tutor should be understood as a pipeline rather than a single AI model.

7.2 Comparison with literature baselines and existing solutions

The proposed architecture is consistent with educational RAG research demonstrating that retrieved educational material can improve grounding.

Levonian et al. showed that RAG can improve educational question-answering but also highlighted the trade-off between strict grounding and natural response preference [2].

The present research extends this concept into a voice-interaction setting and applies it to Sri Lankan O/L Science resources.

Alawwad et al. demonstrated the value of RAG for textbook question answering [3]. The proposed system similarly uses textbook content but differs in its emphasis on a multilingual voice interface and Sri Lankan curriculum resources.

Research on Sinhala ASR demonstrates the importance of multilingual and transfer-learning approaches for Sinhala speech recognition [1]. The present system incorporates Sinhala interaction but does not assume that published Sinhala ASR results directly represent the performance of the selected implementation.

7.3 Research contribution

The research contribution can be summarized as the design of a unified architecture combining:

- curriculum-grounded RAG
- multilingual voice interaction
- local ASR
- semantic vector retrieval
- controlled response generation
- TTS
- educational out-of-scope handling.

The contribution is primarily an applied systems and integration contribution rather than the invention of a new speech-recognition or language model.

7.4 Product contribution

From a product perspective, the Voice Tutor provides a natural spoken interface to the Study_Buddy platform.

Instead of requiring students to type every question, the student can ask questions verbally.

The system can subsequently:

Listen → Understand → Retrieve → Explain → Speak

This interaction model can reduce the interaction barrier for students who are more comfortable asking questions verbally.

The Sinhala capability further improves accessibility for students who may not prefer English-only educational technology.

7.5 Limitations and threats to validity

Despite the proposed architecture, several limitations remain.

Limited formal educational evaluation

A formal controlled study involving O/L students has not yet been completed. Consequently, educational effectiveness cannot currently be quantified.

Speech recognition variability

Speech recognition quality can vary according to:

- Accent.
- Pronunciation.
- Background noise.
- Microphone quality.
- Speaking speed.
- Sinhala-English code switching.

The Whisper Base model also represents a trade-off between computational cost and transcription quality.

Local hardware limitations

Although INT8 CPU inference reduces resource requirements, larger local language models can still introduce latency on low-specification computers.

Retrieval dependency

The quality of the generated answer depends partly on whether the retrieval system successfully identifies the appropriate curriculum chunks.

If the relevant textbook content is not retrieved, the LLM may be unable to produce a complete answer even when the information exists in the knowledge base.

Curriculum coverage

The system's scope is restricted to the indexed curriculum resources. This is intentional but means that information outside those resources may not be answered.

TTS browser dependency

The quality and availability of Sinhala browser speech synthesis depend on the voices supported by the operating system and browser.

Model comparison limitations

The supplied model comparison evaluates three local LLMs using predefined educational prompts and qualitative/score-based criteria. It does not constitute a large-scale standardized benchmark or statistically powered evaluation.

Therefore, the reported scores should be interpreted as comparative development results rather than universal model-performance measurements.

7.6 Implications

The architecture has implications for educational technology development in resource-constrained environments.

A hybrid local/cloud design can reduce recurring operational costs while maintaining access to advanced conversational models where internet connectivity is available.

The local-first speech-processing architecture can also support stronger privacy protection because raw voice recordings do not need to be permanently stored or sent to third-party transcription services.

More broadly, the system demonstrates that curriculum-specific AI assistants can be constructed by combining conventional information-retrieval techniques with modern generative AI rather than relying exclusively on large language models.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research investigated the design and implementation of a syllabus-grounded multilingual conversational AI voice assistant for Sri Lankan G.C.E. O/L Science education.

The research addressed the problem of providing natural voice-based educational interaction while maintaining alignment with prescribed curriculum resources.

A modular architecture was developed combining automatic speech recognition, semantic retrieval, FAISS vector search, retrieval-augmented generation and text-to-speech synthesis.

The knowledge base is constructed from Grade 10 and Grade 11 Science textbooks and Teacher's Guides. This provides an explicit evidence source for conversational responses and distinguishes the proposed system from unrestricted general-purpose conversational AI.

The system also incorporates multilingual interaction, local processing and fallback mechanisms. These design decisions improve operational flexibility and support the project's objective of developing a practical educational technology solution for the Sri Lankan context.

The research demonstrates the technical feasibility of integrating these technologies into a single Voice Tutor component.

However, the effectiveness of the system must ultimately be established through empirical evaluation. In particular, speech-recognition performance, retrieval quality, response quality, latency and user acceptance should be measured using the final evaluation dataset.

8.2 Achievement of objectives

The proposed objectives were addressed as follows.

Objective 1: Develop a local speech-recognition mechanism

Achieved through the integration of faster-whisper, Silero VAD, and CPU-based INT8 inference.

Objective 2: Ground responses in the Sri Lankan Science curriculum

Achieved through the construction of a FAISS vector knowledge base containing curriculum resources and semantic retrieval of relevant content.

Objective 3: Provide conversational explanations

Achieved through integration with an LLM-based reasoning layer.

Objective 4: Support Sinhala and English

Addressed through multilingual response modes and speech synthesis configuration.

Objective 5: Reduce dependence on cloud services

Addressed through local speech recognition and local LLM support.

Objective 6: Identify an appropriate local conversational model

Addressed through comparative evaluation of Mistral 7B, Qwen2.5 7B, and Llama 3.2 3B, with Qwen2.5 7B obtaining the highest overall evaluation score. binuri - model comparison

Objective 7: Provide resilient operation

Addressed through fallback mechanisms for speech recognition, conversational reasoning, and speech synthesis.

8.3 Individual research and product contributions

The individual research contribution is the investigation and implementation of a syllabus-grounded multilingual conversational voice architecture for Sri Lankan school Science education.

The product contribution is a functioning Voice Tutor component capable of receiving voice input, transcribing questions, retrieving curriculum information, generating contextual responses and returning spoken output.

The research therefore contributes both academically and technically by demonstrating how established AI technologies can be combined to address a specific local educational requirement.

8.4 Future directions

Several areas can be investigated in future work.

Improved Sinhala speech recognition

Future work could investigate fine-tuned Sinhala speech-recognition models and larger representative datasets.

Improved retrieval

Future versions could investigate hybrid lexical-semantic retrieval, reranking and more advanced retrieval strategies.

Conversational memory

The assistant could maintain controlled short-term conversation history to support follow-up questions while preventing irrelevant context from influencing answers.

Personalisation

Student performance data could be used to adapt explanations to learner needs.

Visual explanation

The Voice Tutor could be connected more deeply with Study_Buddy 's visual-learning functionality.

Multimodal tutoring

Future versions could combine voice, diagrams, interactive models and textual explanations.

Teacher dashboard

Teachers could receive aggregated insights into commonly requested concepts and student learning difficulties.

Larger empirical evaluation

A larger controlled study involving students and teachers could investigate:

- learning outcomes
- engagement
- usability
- Language Preference
- trust
- curriculum alignment
- learning retention

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GLOSSARY

Term	Definition
ASR	Automatic Speech Recognition; conversion of spoken language into text.
API	Application Programming Interface used for communication between software components.
Embedding	Numerical representation of text that captures semantic meaning.
FAISS	Facebook AI Similarity Search, a library for efficient vector similarity searching.
FastAPI	Python framework used to build the backend API.
LLM	Large Language Model capable of processing and generating natural language.
RAG	Retrieval-Augmented Generation; a technique where relevant external knowledge is retrieved before generating an answer.
STT	Speech-to-Text conversion.
TTS	Text-to-Speech conversion.
VAD	Voice Activity Detection used to identify speech within an audio recording.
Vector Database	A system used to store and search numerical vector representations.
Chunk	A manageable segment of curriculum text stored in the retrieval knowledge base.
Grounding	Restricting generated responses using retrieved evidence or authoritative information.
Hallucination	Generation of information that is unsupported, inaccurate, or unrelated to the available evidence.

INT8	Eight-bit integer numerical representation used to reduce computational and memory requirements.
Cosine Similarity	Similarity measurement between vectors based on the angle between them.
Curriculum Grounding	Restricting AI responses to information supported by the specified educational curriculum.
Whisper	Speech-recognition model developed by OpenAI.
faster-whisper	Optimized implementation of Whisper using CTranslate2.
Ollama	Platform for running large language models locally.
Silero VAD	Voice Activity Detection model used to identify speech regions in audio.

APPENDICES

Appendix A – Research Evaluation Instrument

A.1 Speech-recognition evaluation

Each audio sample should contain a single Science-related question.

For each sample, record:

Table 7 - recognition evaluation

Field	Description
Sample ID	Unique anonymous identifier
Language	English/Sinhala
Question category	Factual/conceptual/comparison/etc.
Reference transcription	Manually verified transcription
ASR output	System transcription
WER	Calculated WER
CER	Calculated CER where applicable
Processing time	Time required for transcription

A.2 Retrieval evaluation

For each question:

Table 8 - Retrieval evaluation

Field	Description
Question ID	Unique identifier
Question	Student/test question
Expected source	Correct textbook/guide
Expected topic	Relevant topic
Retrieved chunk 1	Top result

Retrieved chunk 2	Second result
Retrieved chunk 3	Third result
Retrieved chunk 4	Fourth result
Retrieved chunk 5	Fifth result
Relevance judgement	Relevant/not relevant

A.3 Answer evaluation rubric

Each answer may be evaluated using a 1–5 scale:

Table 9 - Answer evaluation rubric

Criterion	**1**	**3**	**5**
Correctness	Incorrect	Partially correct	Correct
Curriculum relevance	Irrelevant	Partially relevant	Strongly relevant
Explanation	Poor	Adequate	Clear and educational
Grounding	Unsupported	Partially grounded	Clearly grounded
Language	Poor	Understandable	Natural and appropriate

The exact evaluator instructions should be included if human evaluation is conducted.

Appendix A – UI of Component

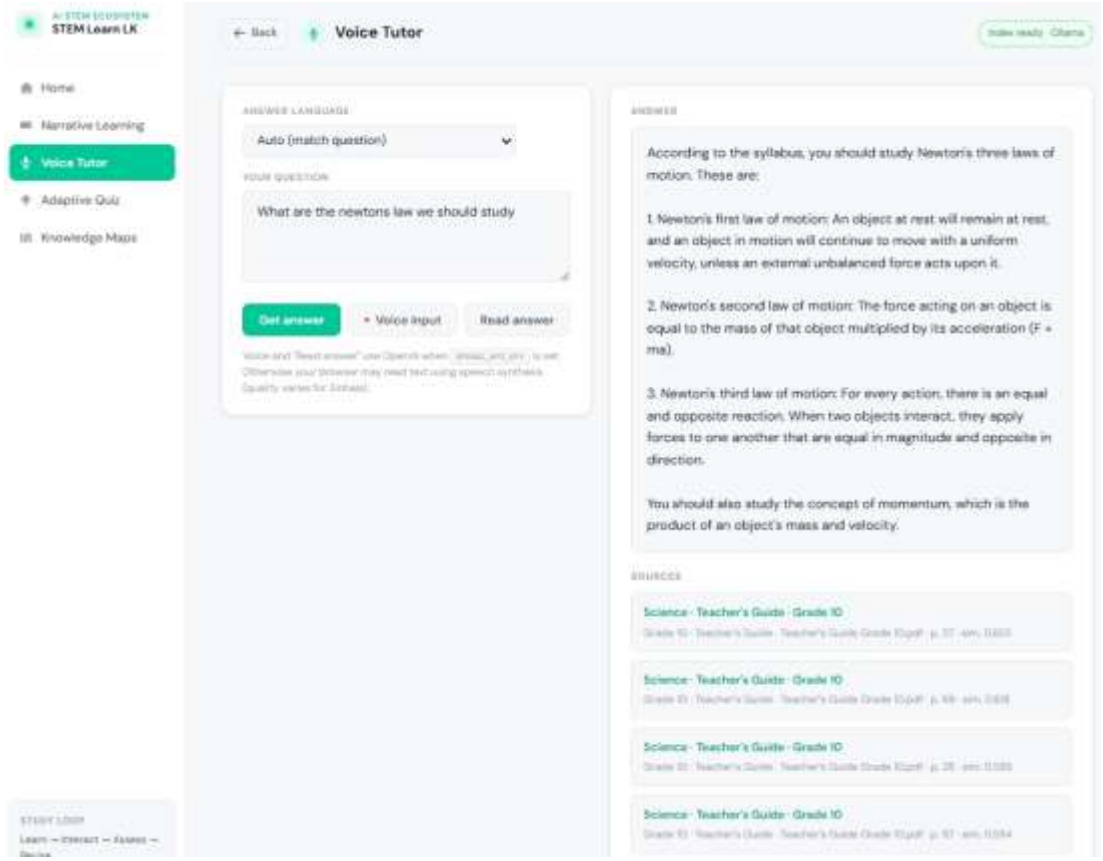


Figure 7 - User Interface 01

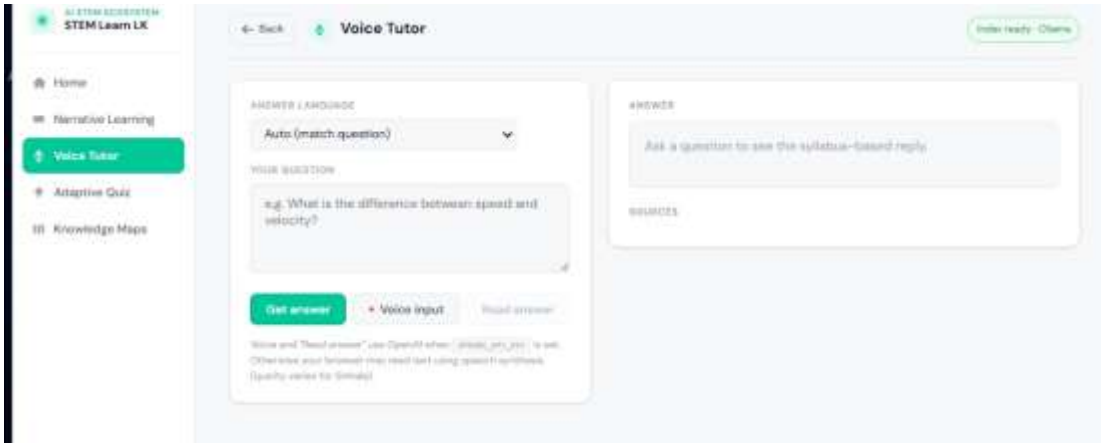


Figure 8 - User Interface 02